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EDUCATION

Université Paris-Saclay , France Ph.D. in Information Theory <i>Graduate School of Computer Science</i>	2022–present
Universidade Estadual de Campinas , Brazil M.Sc. in Applied Mathematics <i>Dissertation: Geometry, Statistics and Applications to Communications and Learning</i>	2021–2022
CentraleSupélec , France Diplôme d'Ingénieur (M.Sc.) <i>Major: Applied Mathematics</i>	2018–2021
Universidade Estadual de Campinas , Brazil B.Sc. in Electrical Engineering <i>Class rank: 1/74, GPA: 0.9196/1.0</i>	2015–2021

RESEARCH INTERESTS

Information theory and communication theory (MSC2020 94A and 94B).

RESEARCH EXPERIENCE

2022–present, *doctoral researcher*, **L2S, CNRS/CentraleSupélec/Université Paris-Saclay**, France.

- Advisor: Prof. Sheng Yang.

2021–2022, *master's researcher*, **IMECC, Universidade Estadual de Campinas**, Brazil.

- Advisor: Prof. Sueli I. R. Costa.

2020–2020, *researcher intern*, **Huawei Technologies**, France.

- Supervisors: Dr. Ingmar Land, Dr. Apostolos Destounis and Prof. Jean-Claude Belfiore.

2019–2020, *student researcher*, **L2S, CNRS/CentraleSupélec/Université Paris-Saclay**, France.

- Supervisor: Prof. Sheng Yang.

2016–2018, *undergraduate student researcher*, **IMECC, Universidade Estadual de Campinas**, Brazil.

- Supervisors: Prof. Henrique N. Sá Earp and Prof. Sueli I. R. Costa.

TEACHING TRAINING

Université Paris-Saclay, France
Certificate “Teaching in Higher Education” (Label “Enseigner dans le Supérieur”) 2022–present

- This program involves training in different pedagogical skills and teaching experience of at least 96h, done in parallel to the preparation of the Ph.D. degree.

TEACHING EXPERIENCE

2022–present, *graduate teaching assistant*, **CentraleSupélec**, France.

- Information theory
- Signal processing
- Communication theory
- Digital signal processing
- Wireless communications

2022–2022, *graduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- Principles of communications I

2021–2021, *undergraduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- Principles of communications I

PREPRINTS

- [Z1] H. K. Miyamoto and S. Yang, “Universal decoding over finite-state additive channels via noise guessing”, *arXiv:2501.12971*, 2025 . doi: 10.48550/arXiv.2501.12971.

JOURNAL PAPERS

- [J4] H. K. Miyamoto, F. C. C. Meneghetti, J. Pinele and S. I. R. Costa, “On closed-form expressions for the Fisher–Rao distance”, *Information Geometry*, vol. 7, pp. 311–354, Nov. 2024, doi: 10.1007/s41884-024-00143-2.
- [J3] H. K. Miyamoto, F. C. C. Meneghetti and S. I. R. Costa, “The Fisher–Rao loss for learning under label noise”, *Information Geometry*, vol. 6, pp. 107–126, Jun. 2023, doi: 10.1007/s41884-022-00076-8.
- [J2] H. K. Miyamoto and S. Yang, “Context-tree-based lossy compression and its application to CSI representation”, *IEEE Transactions on Communications*, vol. 70, no. 7, pp. 4417–4428, Jul. 2022, doi: 10.1109/TCOMM.2022.3173002.
- [J1] H. K. Miyamoto, S. I. R. Costa and H. N. Sá Earp, “Constructive spherical codes by Hopf foliations”, *IEEE Transactions on Information Theory*, vol. 67, no. 12, pp. 7925–7939, Dec. 2021, doi: 10.1109/TIT.2021.3114094.

CONFERENCE PAPERS

- [C6] H. K. Miyamoto and S. Yang, “On universal decoding over discrete additive channels by noise guessing”, *IEEE Information Theory Workshop (ITW)*, pp. 1–6, Sydney, 2025, doi: 10.1109/ITW62417.2025.11240417.
- [C5] H. K. Miyamoto and S. Yang “On universal decoding over memoryless channels with the Krichevsky–Trofimov estimator”, *IEEE International Symposium on Information Theory (ISIT)*, pp. 1498–1503, Athens, 2024, doi: 10.1109/ISIT57864.2024.10619414.
- [C4] F. C. C. Meneghetti, H. K. Miyamoto, S. I. R. Costa and M. H. M. Costa, “Revisiting lattice tiling decomposition and dithered quantisation”, *Geometric Science of Information (GSI)*, Saint-Malo, 2023. In: F. Nielsen and F. Barbaresco (eds.), *GSI 2023*, LNCS 14071, pp. 318–327, Springer, 2023, doi: 10.1007/978-3-031-38271-0_319.

- [C3] F. C. C. Meneghetti, H. K. Miyamoto and S. I. R. Costa, “Information properties of a random variable decomposition through lattices”, *41st International Conference on Bayesian and Maximum Entropy Methods in Science and Engineering (MaxEnt)*, Paris, 2022. In: *Physical Sciences Forum*, vol. 5, no. 1, 2022, doi: 10.3390/psf2022005019.
- [C2] H. K. Miyamoto and S. Yang, “A CSI compression scheme using context trees”, *International Zurich Seminar on Information and Communication (IZS)*, pp. 24–28, Zurich, 2022, doi: 10.3929/ethz-b-000535273.
- [C1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, “Constructive spherical codes in 2^k dimensions”, *IEEE International Symposium on Information Theory (ISIT)*, pp. 1612–1616, Paris, 2019, doi: 10.1109/ISIT.2019.8849464.

DOMESTIC CONFERENCE PAPERS

- [D3] H. K. Miyamoto, “Geometria, estatística e aplicações a comunicações e aprendizado”¹, *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, vol. 10, no. 1, 2023, doi: 10.5540/03.2023.010.01.0004 (in Portuguese).
- [D2] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, “Construção de códigos esféricos usando a fibração de Hopf”, *Jornada Nacional de Iniciação Científica (JNIC)*, 70^a Reunião Anual da SBPC, Maceió, 2018, ISSN: 2176–1221 (in Portuguese).
- [D1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, “Construction of spherical codes using the Hopf fibration”, *XXV Congresso de Iniciação Científica da Unicamp*, Campinas, 2017, doi: 10.19146/pibic-2017-78809.

PATENT APPLICATIONS

- [P1] I. Land, H. K. Miyamoto, A. Destounis and J.-C. Belfiore, “Transfer Learning between Neural Networks”, WO2022207097 (2022), CN117121020 (2023), EP4309079 (2024).

THESES

- [T1] H. K. Miyamoto, “Geometria, Estatística e Aplicações a Comunicações e Aprendizado” (Geometry, Statistics and Applications to Communications and Learning), *M.Sc. dissertation in applied mathematics*, Universidade Estadual de Campinas, Campinas, 2022, doi: 20.500.12733/6633.

SELECTED PRESENTATIONS (WITHOUT PROCEEDINGS)

- IEEE European School on Information Theory (ESIT), *poster presentation*, Eindhoven, The Netherlands, 2024.
- France PhD Workshop on Information Theory, *poster presentation*, Palaiseau, France, 2024.
- IEEE European School on Information Theory (ESIT), *poster presentation*, Bristol, UK, 2023.
- International Conference on Information Geometry for Data Science (IG4DS), *oral presentation*, Hamburg, Germany (virtual), 2022.
- International Congress of Mathematicians (ICM), *oral presentation*, Rio de Janeiro, Brazil, 2018.
- Latin American Week on Coding and Information (LAWCI), *poster presentation*, Campinas, Brazil, 2018.
- Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), *oral presentation*, Curitiba, Brazil, 2017.

¹This is an extended abstract of [T1], published on the occasion of the Clóvis Caesar Gonzaga Award (see Awards).

AWARDS

1st place in Clóvis Caesar Gonzaga Award for master's dissertations in applied and computational mathematics, SBMAC, 2023.

Merit Award in Jornada Nacional de Iniciação Científica (JNIC), SBPC, 2018.

2nd place in Beatriz Neves Award for undergraduate research in applied and computational mathematics, SBMAC, 2018.

Academic Excellence in Undergraduate Work in Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), category Geometry and Topology, PET Matemática/UFPR, 2017.

Scientific Merit in XXV Congresso de Iniciação Científica da Unicamp, PRP/Unicamp, 2017.

Merit Award for the composition produced in Unicamp entrance exam, COMVEST/Unicamp, 2015.

SCHOLARSHIPS

2021–2022, *Master's Degree Scholarship*, São Paulo Research Foundation (FAPESP), 2021/04516–8.

2018–2020, *Eiffel Excellence Scholarship*, French Ministry of Europe and Foreign Affairs.

2016–2018, *Scientific Initiation Scholarship*, São Paulo Research Foundation (FAPESP), 2016/05126–0.

2013–2013, *Junior Scientific Initiation Scholarship*, Brazilian National Council for Scientific and Technological Development (CNPq).

ACADEMIC SERVICE

Scientific societies membership

2020–present, *student member*, Sociedade Brasileira para o Progresso da Ciência (SBPC).

2018–present, *graduate student member*, Institute of Electrical and Electronics Engineers (IEEE).

- Information Theory Society, Communications Society, Signal Processing Society.

Reviewing

2025, *reviewer*, Information Geometry, Springer.

2023–2025, *reviewer*, IEEE International Symposium on Information Theory (ISIT).

2023, *reviewer*, Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT).

2022, *reviewer*, Journal of Communication and Information Systems (JCIS).

Committees and representative positions

2023–present, *doctoral and postdoc researchers representative*, Laboratory of Signals and Systems (L2S).

2023, *organising committee*, Encontro de Códigos, Reticulados e Informação, IMECC, Unicamp.

2018–2019, *student representative*, Departamento de Comunicações, FEEC, Unicamp.

LANGUAGES

Portuguese	native
English	advanced (Cambridge CAE)
French	advanced
Spanish	intermediate
Japanese	elementary (JLPT N4)

Last updated: November 21, 2025